Chidera Onyebu

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Backend and systems engineer: a deployed payments marketplace with a double-entry ledger, an independent verifier for an AI-commerce evidence standard, and an LLM agent whose safety gate lives in the graph, not the prompt.

Education

University of Maryland, College Park — College Park, MD

B.S. Computer Science and B.S. Applied Mathematics — GPA 3.5/4.0 — *Aug. 2024 – May 2028*

- **Honors: Three-time Gold Medalist, Nigerian National Mathematics Olympiad**; 1st Place (Senior Category), Rotary Mathematics Competition, 2024; Alpha Lambda Delta.
- **Coursework:** Algorithms, Data Structures, Computer Systems (C, MIPS), Organization of Programming Languages (OCaml, Rust), Probability & Statistics, Linear Algebra.

Projects

ToolBelt — Two-Sided Gig Marketplace | *FastAPI, PostgreSQL, TypeScript, Stripe, Docker* — Jul. 2026 – Present

- Built and deployed a two-sided marketplace to a public domain: a 7,800-line FastAPI/PostgreSQL REST API and a 4,300-line TypeScript Expo client on iOS, Android, and web, with Twilio phone-verified signup over TLS.
- Eliminated a double-charge defect in which Stripe's 24-hour idempotency-key expiry turned a retried refund into a second real charge, by journaling every money-moving call on a separate connection *before* the call is issued.
- Implemented and validated a double-entry ledger across authorize, capture, and payout against live Stripe test mode at a zero trial balance, with a reconciliation sweeper that is read-only by construction.

TradeDesk — LLM Agent Platform | *Python, LangGraph, FastAPI, MCP, pytest* — Jun. – Aug. 2026

- Made an unconfirmed trade **structurally impossible**: the compiled LangGraph state machine contains no trade-to-confirmation edge and a test asserts its absence, so no prompt or handler change can bypass it.
- Proved the gate under attack: 18 adversarial evaluations (prompt injection, privilege escalation, scope) asserted at the broker through a spy pass **18/18 with 0 unauthorized executions**, across **293 tests**.

aep-verify — AI-Commerce Evidence Verifier | *Python, Ed25519, RFC 8785* — Oct. 2026 – Present

- Built a verifier for the A-Comm Evidence Protocol, an open draft for proving what an AI agent bought, from the spec alone, passing all **55** conformance vectors under **250 tests**.
- Attacked one disputed $55 purchase **22 ways**: **5** attacks passed, **2** byte-identical to the honest bundle. Per-event head logging cut that to **2**. Filed **9** spec issues upstream.

Compilers, Runtimes, and Systems Programming | *C, OCaml, Rust* — 2025 – 2026

- Implemented the recursive executor for a UNIX shell in C, walking the parse tree to run pipelines, `&&` chains, and subshells over `fork`/`exec` with `dup2` redirection, exit-status propagation, and ambiguous-redirect detection.
- Built a front-to-back toolchain for a C subset in OCaml (1,700+ lines) — lexer, parser, interpreter, static type checker, stack-machine compiler — plus regex-to-NFA-to-DFA construction validated by a 100-case property test.
- Wrote three garbage collectors in Rust over one shared heap and reachability pass: reference-counting, mark-and-sweep, and semispace stop-and-copy that relocates live objects and rewrites both child pointers and stack roots.

Experience

Capstone Engineer, AI.Accelerate FY26 — Jun. – Aug. 2026

ServiceNow (external program) — *Remote*

- Built and demoed TradeDesk, a self-proposed agentic trading assistant, live at Discover ServiceNow as the capstone of a 12-session program, backed by an 8-risk safety audit in which every claimed control cites a named passing test.

Machine Learning Contributor, AI4ALL — Jun. – Aug. 2026

Apply AI Program — 4-person team — *Remote*

- Shipped the deployed inference app for a 4-person image-model team and merged it upstream, then caught a capture-session leak their own check missed: **94.7% of test examples** shared a session with training data, inflating accuracy by **3.3 points**.

Peer Mentor and Academic Mentor — Sept. 2024 – Present

ColorStack & Alpha Lambda Delta Honor Society, UMD — *College Park, MD*

- Coach 50+ students one-on-one and lead 18+ debugging workshops and review sessions, teaching problem decomposition rather than supplying answers.

Technical Skills

Languages: Python, TypeScript/JavaScript, Java, C, OCaml, Rust, SQL, HTML/CSS
Frameworks: FastAPI, REST APIs, LangGraph, React, React Native/Expo, Node.js, Pydantic, SQLAlchemy, pytest, Jest
Infrastructure: PostgreSQL, Redis, Docker, Stripe, Render, Cloudflare, GitHub Actions CI/CD, Git, UNIX
Practices: `mypy --strict`, unit, integration, and property-based testing, schema migrations, idempotency design, AI-assisted development (Claude Code)